

Fig. 10: Microstructure of CNT

resistance of the forest.

When pressure is applied to the membrane, a strain is generated in it as well as in the forest, leading to a change in the resistance of the forest due to its piezoresistive effect. The source of piezoresistive effect of the CNT forest is related to the change in distances between individual CNTs due to lateral strains applied to the forest.

As shown in Fig. 9, a positive pressure that deflects the membrane downwards can widen the CNT separation around the centre region of the membrane and increase the resistance in the region. However, it can also narrow the separation around the two border regions between the membrane and the substrate, thereby decreasing resistance in these regions. The opposite behaviour can be caused by a negative pressure.

The polarity may be determined by the collective piezoresistive effects across the forest including the two border regions. However, these border regions are relatively large and play the dominant role in the overall piezoresistive effect.

Fig. 10 shows the individual CNT and needle-like microstructure

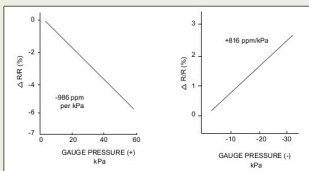


Fig. 11: Forest resistance against gauge pressure

of the CNT forest. The fabrication process of the CNT forest starts with formation of a Si-nitride mask on both sides of a Si substrate, followed by patterning of a square window in the mask on the backside of the substrate. A molybdenum film is deposited with an electron beam evaporation technique to form the metal pads on the front side of the substrate. A catalyst layer made of 2nm Fe on 10nm Al_2O_3 is deposited in the region where the CNT forest will be grown.

Chemical vapour deposition technique is used to grow a forest of 400-600μm height using C_2H_2 as the carbon source, followed by deposition of 8μm thick parylene-C film binding the tops of individual CNTs. The backside window is made by dry etching of the silicon substrate. Fig. 11 shows the responses of the sensor for positive and negative gauge pressures.

The author retired from Metallurgical & Engineering Consultants (I) Ltd as deputy general manager, I/C

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